

COLLEGE STATION EASTWOOD, TX, USA

WMO: 747460

Lat: 30.589N

Lon: 96.365W

Elev: 93

StdP: 100.21

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	-0.1	-9.8	1.6	3.8	-7.5	2.0	5.4	10.9	13.0	9.6	12.8	3.5	350	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	37.7	24.2	36.7	24.2	35.6	24.3	26.6	32.4	26.0	31.6	25.7	31.1	3.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.6	28.1	24.8	20.1	27.8	24.3	19.5	27.5	83.0	32.0	80.9	31.8	79.1	31.2	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.4	7.6	DB	-5.1	39.6	2.0	1.7	-6.5	40.9	-7.7	41.9	-8.9	42.8	-10.3	44.1	
			WB	-6.3	27.0	1.9	0.6	-7.7	27.5	-8.8	27.8	-9.9	28.2	-11.3	28.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.8	10.9	13.1	16.6	20.5	24.7	28.1	29.4	29.7	26.9	21.7	16.0	11.8
	DBStd	7.86	5.51	5.72	4.98	3.90	3.04	1.99	1.61	1.82	2.85	4.19	5.05	5.37
	HDD10.0	146	55	31	9	1	0	0	0	0	0	0	9	41
	HDD18.3	848	236	162	92	23	2	0	0	0	1	18	101	213
	CDD10.0	4098	84	117	212	314	456	543	602	611	507	364	191	97
	CDD18.3	1758	7	14	37	87	199	293	344	353	257	124	32	11
	CDH23.3	18707	26	71	203	639	1842	3311	4249	4393	2684	1049	199	41
	CDH26.7	8867	1	12	28	154	719	1617	2267	2399	1280	357	29	3
(14) Wind	WSAvg	3.5	3.7	4.0	4.1	4.2	3.9	3.4	3.1	2.9	2.8	3.1	3.4	3.5
(15) Precipitation	PrecAvg	1040	82	73	69	86	121	100	58	58	116	104	82	71
	PrecMax	1550	396	249	154	223	289	321	179	270	261	328	236	220
	PrecMin	727	6	2	7	2	14	2	0	2	13	9	5	6
	PrecStd	240	74	54	40	57	73	86	46	54	72	74	54	47
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	28.1	29.2	32.4	35.0	38.0	38.6	39.5	38.0	33.5	29.2	26.8
		MCWB	18.1	18.3	19.8	22.4	24.0	24.3	23.8	23.7	23.0	22.7	21.6	20.2
	2%	DB	23.7	25.4	27.3	29.9	33.3	36.1	37.4	37.9	35.9	31.9	27.5	24.2
		MCWB	17.6	18.2	19.6	21.4	23.7	23.9	24.5	24.1	23.5	22.9	20.8	19.0
	5%	DB	21.9	23.4	25.8	28.3	32.1	34.6	36.3	36.9	34.1	30.4	25.8	22.6
		MCWB	17.5	17.8	19.0	20.8	23.4	24.3	24.5	24.3	23.3	22.1	19.7	18.7
	10%	DB	20.1	21.7	23.9	27.1	30.7	33.4	35.0	35.6	32.9	28.7	23.9	21.0
		MCWB	16.3	17.5	18.2	20.2	23.0	24.3	24.5	24.4	23.4	21.2	19.5	17.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.8	21.3	22.2	24.4	26.4	27.1	27.4	26.8	26.6	25.6	23.7	21.9
		MCDB	22.7	24.4	26.0	28.6	31.6	33.4	34.9	32.2	31.3	29.5	26.7	24.1
	2%	WB	19.7	20.4	21.3	23.3	25.2	26.3	26.4	26.2	25.8	24.8	22.6	20.8
		MCDB	22.3	23.4	24.7	27.3	30.3	32.0	33.0	31.8	30.4	28.6	25.2	23.2
	5%	WB	18.8	19.5	20.5	22.5	24.6	25.7	25.8	25.8	25.4	24.1	21.6	19.7
		MCDB	21.2	22.1	23.7	26.4	29.4	30.9	31.9	31.4	30.0	27.7	24.3	22.0
	10%	WB	17.2	18.5	19.7	21.8	24.0	25.2	25.4	25.5	24.9	23.4	20.4	18.3
		MCDB	19.4	21.1	22.7	25.4	28.5	30.2	31.3	31.2	29.5	26.7	23.3	20.4
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	10.7	10.9	11.1	10.5	10.2	11.0	11.4	11.0	11.4	11.1	10.8
		MCDBR	12.6	12.1	11.4	11.4	11.4	11.8	12.7	13.2	12.6	11.9	12.0	11.8
	5% WB	MCWBR	7.9	7.0	5.6	5.2	3.8	2.9	2.8	2.8	3.4	4.8	6.5	7.7
		MCWBR	10.3	9.8	9.5	9.4	9.7	9.8	11.0	11.1	10.2	9.6	10.1	10.2
(40) Clear-Sky Solar Irradiance	taub		0.326	0.345	0.368	0.416	0.465	0.450	0.467	0.457	0.425	0.374	0.367	0.327
	taud		2.503	2.461	2.398	2.275	2.181	2.276	2.257	2.281	2.326	2.450	2.408	2.510
	Ebn at Noon		906	918	916	878	834	842	825	832	848	874	847	886
	Edh at Noon		89	101	115	135	150	135	137	133	122	101	97	84
(44) All-Sky Solar Radiation	RadAvg		2.89	3.26	4.40	5.36	5.79	6.38	6.47	6.10	5.14	4.25	3.16	2.53
	RadStd		0.28	0.62	0.41	0.39	0.44	0.48	0.48	0.35	0.46	0.56	0.35	0.32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+98

Nomenclature: See separate page